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Media and Political Persuasion: Evidence from Russia

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1 Big picture

- **Question.** Can an independent national television channel persuade voters and change election outcomes when most other national media are government controlled?
- **Setting.** The paper studies Russia's December 1999 parliamentary election. The new pro-government party Unity was strongly promoted by state television, while NTV was the only national television channel with editorial independence from the government.
- **Treatment.** Local availability of NTV broadcasts. NTV reached roughly three-quarters of the Russian population, but the signal was not available everywhere.
- **Core empirical idea.** Variation in NTV availability came from transmitter locations and signal propagation. Conditional on observables and region fixed effects, this variation is used to compare otherwise similar localities with and without access to the independent channel.
- **Main aggregate result.** NTV availability reduced the vote share of the pro-government Unity party, increased support for opposition parties covered favourably by NTV, and reduced turnout.
- **Magnitude.** The paper's preferred calculations imply that NTV broadcasts decreased the aggregate vote for Unity by about 8.9 percentage points, increased the combined vote for the major opposition parties by about 6.3 percentage points, and decreased turnout by about 3.8 percentage points.
- **Voter-count interpretation.** In a sample with 88.4 million registered voters, the estimates imply roughly 5.7 million fewer votes for Unity, 2.8 million more votes for OVR/SPS/Yabloko, and 3.4 million fewer voters turning out.
- **Individual-level result.** Survey evidence shows that NTV signal strength predicts self-reported watching of NTV. Instrumented NTV watching increases reported votes for opposition parties even after controlling for pre-election voting intentions.
- **Persuasion result.** The positive message of NTV has a persuasion rate close to that found in U.S. media studies, while the negative message against Unity has a much larger persuasion rate.
- **Long-run result.** Some effects of NTV exposure persist into the 2003 election, after the independent channel had been taken over and its editorial line had changed.
- **Economic takeaway.** Independent media can have very large persuasion effects in weakly institutionalized democracies, especially when voters have weak priors, party identities are unstable, and government-controlled media dominate the information environment.

2 Incremental contribution of the paper

2.1 What this paper adds relative to the media-persuasion literature

- Prior evidence on media persuasion often comes from stable democracies, such as U.S. newspapers, radio, or cable news, where party identities and ideological positions are relatively well known.
- This paper studies a setting with weak democratic institutions, unstable party systems, low voter information, and a highly concentrated media environment.
- The paper therefore shows that media persuasion can be much larger when voters have weak prior beliefs and when one-sided political coverage is the norm.
- It also studies an *independent* media outlet in an environment dominated by state media, rather than a partisan outlet in a competitive media market.

Interpretation: The incremental contribution is not simply estimating another media effect. The paper identifies a setting in which media influence is expected to be large and documents that it is indeed large.

2.2 What this paper adds methodologically

- The paper exploits signal availability rather than voluntary media consumption alone.
- It uses transmitter locations and signal propagation to measure the geographic reach of NTV.
- It controls for city status, population, wages, socioeconomic characteristics, 1995 electoral outcomes, and region fixed effects.
- It adds a placebo exercise using 1995 outcomes, when NTV had negligible audience, to test whether NTV availability is merely proxying for pre-existing political preferences.
- It combines aggregate official election results with individual survey data and an IV strategy based on signal strength.

Interpretation: The paper’s empirical design is valuable because it moves beyond simple correlations between media consumption and voting. It uses quasi-geographic variation in access to media and checks whether that variation predicts pre-treatment outcomes.

2.3 What this paper adds conceptually

- The paper distinguishes between positive persuasion and negative persuasion in a multiparty election.
- NTV’s positive message was to support OVR and liberal parties; its negative message was not to vote for Unity.
- The negative-message persuasion rate is much larger than the positive-message persuasion rate.
- This distinction is especially important in multiparty systems, because voters may respond by switching to another party, abstaining, or becoming undecided.

Interpretation: The paper expands persuasion analysis beyond a two-party framework. It shows that media can affect both vote choice and participation, and that negative messages can be especially powerful.

2.4 What this paper adds to political economy of media control

- The paper provides direct evidence that independent media can discipline government-controlled narratives.
- It quantifies the electoral cost to the pro-government party from access to independent television.
- It shows that control of national television can be politically valuable in young democracies.
- The findings help explain why governments in weak democracies have strong incentives to control television networks.

Interpretation: The contribution is closely linked to authoritarian and hybrid-regime politics: media independence is not only a normative issue but also a quantitatively important determinant of political power.

3 Data and measurement

3.1 Political and media context

- In August 1999, Vladimir Putin was appointed prime minister while his popularity was initially very low.
- The pro-government party Unity was created shortly before the December 1999 parliamentary election.
- State-controlled national television strongly supported Unity and attacked its rivals.
- NTV was the only national television channel independent of the government.

- NTV's political coverage opposed Unity and was more favorable to OVR, SPS, and Yabloko.

Interpretation: This was an environment in which television was the primary source of political information for many voters and the party system was unstable. Media coverage could therefore plausibly affect both information and political preferences.

3.2 NTV availability and transmitter data

- The key treatment variable is the predicted availability of NTV in 1999 at the subregional level.
- NTV inherited part of its transmitter infrastructure from a Soviet educational channel.
- Transmitter placement was related to city size and economic characteristics, so the paper controls carefully for these variables.
- Availability is constructed from information on transmitter locations, technical signal reach, and geographic signal propagation.
- The empirical analysis uses NTV signal power and predicted NTV availability as measures of access to independent television.

Interpretation: The identifying variation is geographic. Some localities could watch NTV and others could not, even within the same broader region.

3.3 Aggregate election data

- The main aggregate sample uses official election results at the subregional level.
- The baseline 1999 sample includes 2,005 subregions in 81 regions.
- Outcomes are measured as percentages of the total vote, except turnout, which is measured relative to registered voters.
- Key electoral outcomes include votes for Unity, OVR, SPS, Yabloko, KPRF, LDPR, and turnout.
- The paper also uses 1995 and 2003 elections for placebo and persistence tests.

Interpretation: The aggregate data make it possible to estimate the electoral consequences of NTV access across the entire Russian Federation, not only among survey respondents.

3.4 Political parties and interpretation of media content

- **Unity:** the main pro-government party in 1999; opposed by NTV and supported by state television.
- **OVR:** a centrist opposition party supported by NTV and opposed by state television.
- **SPS and Yabloko:** liberal parties covered more favorably by NTV than by state television.
- **KPRF and LDPR:** parties with relatively similar coverage by NTV and state television.

Interpretation: This categorization gives directional predictions. NTV should reduce support for Unity, raise support for OVR/SPS/Yabloko, and have weaker or ambiguous effects on KPRF and LDPR.

3.5 Survey data

- The individual-level analysis uses a postelection survey with information on media consumption and reported vote.
- It also uses a pre-election survey that asked about voting intentions about a month before the election.
- The first-stage variable is whether the respondent reported watching NTV.
- The instrument is NTV signal power in the respondent’s location.
- Controls include demographics, education, marital status, a consumption index, local population, wages, and pre-election voting intentions in some specifications.

Interpretation: The survey data allow the paper to connect NTV signal access to actual watching and to test whether NTV changed voter choices relative to their prior intentions.

4 Empirical specifications and equations

4.1 Baseline aggregate cross-section

The main aggregate specification can be written as

$$Y_{sr} = \alpha + \beta NTV_{sr} + X'_{sr}\gamma + \delta_r + \varepsilon_{sr}, \quad (1)$$

where Y_{sr} is an electoral outcome in subregion s of region r , NTV_{sr} is predicted NTV availability in 1999, X_{sr} includes political and socioeconomic controls, and δ_r are region fixed effects.

Interpretation: The coefficient β compares subregions with different NTV access within the same region, conditional on city status, population and wage polynomials, 1995 vote shares, and other socioeconomic controls.

4.2 Placebo regression for 1995

The placebo specification uses the same treatment variable but replaces the outcome with 1995 election results:

$$Y_{sr,1995} = \alpha + \beta NTV_{sr,1999} + X'_{sr}\gamma + \delta_r + \varepsilon_{sr}. \quad (2)$$

Interpretation: If NTV availability in 1999 is merely proxying for persistent political preferences, it should predict 1995 outcomes. Table 4 shows that it generally does not.

4.3 Aggregate panel specification

The panel specification compares outcomes across 1995 and 1999:

$$Y_{srt} = \alpha_s + \lambda_t + \beta (NTV_{sr} \times Post1999_t) + \varepsilon_{srt}. \quad (3)$$

Interpretation: Subregion fixed effects absorb time-invariant local political preferences. The key variation comes from whether subregions with NTV access changed differently between 1995 and 1999.

4.4 Long-term effect in 2003

To assess persistence, the paper estimates regressions of the form

$$Y_{sr,2003} = \alpha + \beta NTV_{sr,1999} + X'_{sr}\gamma + \delta_r + \varepsilon_{sr}. \quad (4)$$

Interpretation: This asks whether the political effects of the independent NTV period remain visible after NTV was taken over and after the political environment changed.

4.5 Individual-level first stage

The first-stage relationship is

$$WatchNTV_i = \pi_0 + \pi_1 SignalPower_i + Z_i' \pi_2 + u_i, \quad (5)$$

where $WatchNTV_i$ is a self-reported indicator for watching NTV and $SignalPower_i$ is the technical signal strength in the respondent's location.

Interpretation: A strong positive π_1 validates signal power as an instrument for exposure to NTV programs.

4.6 Individual-level second stage

The second-stage relationship can be written generically as

$$Vote_{ip} = F\left(\alpha_p + \beta_p \widehat{WatchNTV}_i + Z_i' \gamma_p\right), \quad (6)$$

where $F(\cdot)$ is a probit or IV-probit link function and p indexes political outcomes such as voting for Unity, OVR, SPS, Yabloko, KPRF, LDPR, or turnout.

Interpretation: The individual-level specification asks whether people induced to watch NTV by signal access reported different voting choices.

4.7 Persuasion rate of a positive message

For parties supported by NTV, the persuasion rate is based on the idea that only those who would not otherwise vote for supported parties can be persuaded. Let v be the vote share of supported parties, t turnout, and e exposure to NTV. The paper's continuous-exposure logic implies a persuasion rate of the form

$$f^+ = \frac{1}{1 - tv_0} \left[t \frac{dv}{de} + v_0 \frac{dt}{de} \right]. \quad (7)$$

Interpretation: This formula adjusts the effect on vote shares for both the pool of persuadable voters and changes in turnout. The positive-message persuasion rate is about 7.7 percent.

4.8 Persuasion rate of a negative message

For the negative message against Unity, only potential Unity voters are persuadable. Let μ be the Unity vote share. The corresponding persuasion rate is

$$f^- = \frac{1}{t\mu_0} \left[-t \frac{d\mu}{de} - \mu_0 \frac{dt}{de} \right]. \quad (8)$$

Interpretation: The negative-message persuasion rate is much larger, around 65 percent, indicating that NTV was especially effective at discouraging voters from supporting Unity.

5 Identification and empirical design

5.1 Core identifying assumption

- Conditional on controls and region fixed effects, NTV availability is not correlated with unobserved political preferences except through NTV exposure.

- The paper recognizes that transmitter locations are not random; they are related to city size and economic development.
- To address this, the paper includes city indicators, high-order polynomials in population and average wages, socioeconomic controls, and 1995 electoral controls.

Interpretation: The identification does not rely on unconditional randomness of transmitters. It relies on conditional quasi-random variation in availability after controlling for observable determinants.

5.2 Why 1995 electoral controls matter

- 1995 outcomes proxy for pre-existing local political preferences.
- If NTV availability is correlated with liberal, nationalist, communist, or pro-government preferences, these correlations should appear in 1995 vote shares.
- Including 1995 outcomes reduces the concern that NTV availability simply reflects political geography.

Interpretation: The controls make the 1999 estimates closer to a comparison of subregions with similar past electoral behavior but different access to NTV.

5.3 Why the placebo test is important

- NTV's 1999 availability should not affect 1995 outcomes because NTV had negligible audience at that time.
- The placebo regressions test whether NTV availability predicts 1995 electoral outcomes using the same specification as the main regressions.
- The lack of strong placebo effects supports the exclusion restriction.

Interpretation: The placebo design is one of the strongest checks that the main results are not driven by latent political leanings of areas receiving NTV signal.

5.4 Why the individual-level IV matters

- Aggregate estimates could in principle be affected by ecological inference concerns.
- Survey data directly link signal power to watching NTV and watching NTV to self-reported vote.
- The first stage shows that signal strength strongly predicts NTV watching.
- The second stage shows that NTV exposure changes individual reported vote choices.

Interpretation: The individual-level analysis provides a microfoundation for the aggregate estimates.

5.5 Main limitations

- NTV availability is not experimentally assigned.
- Signal availability may still be correlated with unobserved local characteristics despite extensive controls.
- Survey measures of media exposure and voting are self-reported and may contain recall error.
- The paper observes one politically important election in a distinctive institutional environment, so external validity depends on how similar other settings are.

Interpretation: The paper is most convincing as evidence for large media effects in weakly institutionalized political environments with concentrated media markets.

6 Tables and figures

The original paper contains no main plotted figures in the printed article. The central evidence is in the tables below, which are directly cropped from the paper.

6.1 Table 1: Correlates of NTV transmitter locations

Read:

- Transmitter location is correlated with large-city status, population, and wages in simple specifications.
- Once high-order population and wage controls are included, 1995 political outcomes and socioeconomic characteristics are not jointly strong predictors of transmitter locations.
- NTV transmitters in 1997 strongly predict transmitters in 1999.

Economic message: Table 1 motivates the control strategy. Transmitters are not randomly placed, but the main observables that predict them can be explicitly controlled for.

TABLE 1—CORRELATES OF THE LOCATION OF NTV TRANSMITTERS IN 1999						
	NTV Transmitters in 1999					
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Determinants of transmitter location</i>						
City dummy, 1998	0.1562 [0.0271]***	0.0633 [0.0317]**		0.0484 [0.0370]	0.0496 [0.0372]	0.04 [0.0234]*
Population, 1998	0.001 [0.0001]***					
Average wage, 1998	0.0732 [0.0335]**					
Fifth-order polynomial of log population, 1998 (<i>F</i> -statistic, population polynomial)		Yes (24.49)***		Yes (25.05)***	Yes (24.83)***	Yes (9.69)***
Fifth-order polynomial of log average wage, 1998 (<i>F</i> -statistic, wage polynomial)		Yes (10.65)***		Yes (3.12)**	Yes (3.03)**	Yes (5.3)***
<i>Election results, 1995</i>						
Vote for NDR (progovernment) in 1995, percent			0.0014 [0.0013]	−0.0012 [0.0014]	−0.0015 [0.0014]	−0.0005 [0.0012]
Democratic Russia's Choice (Liberal), percent			0.0099 [0.0110]	−0.0038 [0.0026]	−0.0043 [0.0026]*	−0.0022 [0.0016]
Vote for Yabloko (Liberal) in 1995, percent			0.035 [0.0089]***	−0.0062 [0.0098]	−0.0082 [0.0099]	−0.0075 [0.0057]
Vote for KPRF (Communist) in 1995, percent			−0.0002 [0.0015]	−0.0001 [0.0016]	−0.0002 [0.0015]	−0.0008 [0.0009]
Vote for LDPR (Nationalist) in 1995, percent			−0.001 [0.0028]	−0.0023 [0.0025]	−0.0025 [0.0025]	−0.0016 [0.0020]
Voter turnout in 1995, percent			−0.0084 [0.0018]***	−0.0004 [0.0020]	−0.0002 [0.0020]	0.0001 [0.0016]
<i>Socioeconomic characteristics, 1998</i>						
Doctors per 1,000, 1998					0.0016 [0.0012]	−0.00002 [0.0008]
Nurses per 1,000, 1998					−0.0005 [0.0004]	−0.0003 [0.0002]
Population change, 1998					−0.0076 [0.0048]	−0.0014 [0.0017]
Migration rate, 1998					0.0082 [0.0096]	−0.0017 [0.0074]
Fraction of retired people, 1998					−0.0012 [0.0019]	−0.0009 [0.0010]
Fraction of unemployed, 1998					0.0078 [0.0076]	−0.0011 [0.0045]
Fraction of population employed in farms, 1998					0.0073 [0.0061]	−0.0035 [0.0021]
Crime rate, 1998					0.0001 [0.0001]	0.0001 [0.0001]
NTV transmitters in 1997						0.7227 [0.0342]***
Region fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,058	2,058	1,948	1,727	1,723	1,723
<i>R</i> ²	0.34	0.38	0.29	0.39	0.39	0.71
<i>F</i> -statistic, determinants of transmitter location	(87.78)***	(38.7)***		(10.58)***	(10.51)***	(5.62)***
<i>F</i> -statistic, election results variables			(23.54)***	(0.86)	(1.29)	(0.63)
<i>F</i> -statistic, socioeconomic characteristics					(1.15)	(1.19)

Notes: Robust standard errors in brackets. *F*-statistics in parentheses.

***Significant at the 1 percent level.

**Significant at the 5 percent level.

6.2 Table 2: Main aggregate effect on 1999 voting behavior

Read:

- NTV availability reduces the Unity vote share by about 15.5 percentage points in the controlled specification.
- It increases votes for OVR, SPS, and Yabloko.

- It has weaker or mixed effects on KPRF and LDPR, which were covered more similarly by NTV and state television.
- It reduces turnout by about 6.7 percentage points in the controlled specification.

Economic message: Access to independent television shifted voters away from the pro-government party and toward parties receiving more favorable NTV coverage. It also discouraged some voters from participating.

6.3 Table 3: Aggregate panel evidence

Read:

- The panel compares 1995 and 1999 outcomes with subregion fixed effects.
- NTV availability is positively associated with SPS and Yabloko outcomes and not with the main parties that had similar media coverage.
- The panel cannot estimate effects on Unity and OVR because these parties were not present in the same form in 1995.

Economic message: The panel analysis supports the interpretation that NTV affected parties whose media coverage changed, rather than simply reflecting stable local political preferences.

TABLE 2—EFFECT OF NTV AVAILABILITY ON VOTING BEHAVIOR IN 1999, AGGREGATE DATA, CROSS SECTION

	Opposed by NTV in 1999; vote for Unity in 1999 (centrist, progovernment)		Supported by NTV in 1999; vote for OVR in 1999 (centrist, opposition)	
<i>Panel A</i>				
NTV availability in 1999	-17.72 [2.51]***	-15.48 [2.77]***	5.72 [1.95]***	3.62 [1.72]**
Electoral controls from 1995	No	Yes	No	Yes
Socioeconomic controls, 1998	Yes	Yes	Yes	Yes
Regional dummies	Yes	Yes	Yes	Yes
Observations	2,005	1,686	2,005	1,686
Number of regions	81	79	81	79
R ²	0.67	0.66	0.79	0.82
Supported by NTV in 1999				
	Vote for SPS in 1999 (liberal)		Vote for Yabloko in 1999 (liberal)	
<i>Panel B</i>				
NTV availability in 1999	4.47 [1.07]***	3.52 [1.15]***	4.58 [0.84]***	3.85 [0.67]***
Electoral controls from 1995	No	Yes	No	Yes
Socioeconomic controls, 1998	Yes	Yes	Yes	Yes
Regional dummies	Yes	Yes	Yes	Yes
Observations	2,005	1,686	2,005	1,686
Number of regions	81	79	81	79
R ²	0.78	0.82	0.78	0.84
Similar coverage by NTV and state TV in 1999				
	Vote for KPRF in 1999 (communist)		Vote for LDPR in 1999 (nationalist)	
<i>Panel C</i>				
NTV availability in 1999	1.68 [2.03]	3.92 [1.86]**	-1.72 [0.66]**	-1.39 [0.61]**
Electoral controls from 1995	No	Yes	No	Yes
Socioeconomic controls, 1998	Yes	Yes	Yes	Yes
Regional dummies	Yes	Yes	Yes	Yes
Observations	2,005	1,686	2,005	1,686
Number of regions	81	79	81	79
R ²	0.74	0.81	0.70	0.73
Voter turnout in 1999				
<i>Panel D</i>				
NTV availability in 1999	-6.54 [1.94]***	-6.67 [1.42]***		
Electoral controls from 1995	No	Yes		
Socioeconomic controls, 1998	Yes	Yes		
Regional dummies	Yes	Yes		
Observations	2,005	1,686		
Number of regions	81	79		
R ²	0.71	0.81		

Notes: All dependent variables are measured in percentages of total vote. Electoral controls include the results of Duma elections in December 1995, in particular, vote for KPRF (communists), vote for Yabloko, vote for NDR (Our Home is Russia), vote for LDPR (Liberal-Democratic Party of Russia), vote for Democratic Russia's Choice, and voter turnout. The set of socioeconomic controls includes fifth-order polynomial of population, fifth-order polynomial of average wage, number of nurses per capita, number of doctors per capita, and dummy for cities. Robust standard errors adjusted for clusters by region in brackets.

***Significant at the 1 percent level.
**Significant at the 5 percent level.

TABLE 3—EFFECT OF NTV AVAILABILITY ON VOTING BEHAVIOR, AGGREGATE PANEL DATA

	Supported by NTV in 1999		Similar coverage by NTV and state TV in 1999		
	Vote for SPS and its predecessor Democratic Russia's Choice (liberal)	Vote for Yabloko (liberal)	Vote for KPRF (communist)	Vote for LDPR (nationalist)	Voter turnout
NTV availability	6.65 [1.40]***	1.84 [0.76]**	-2.2 [2.12]	1.18 [1.38]	-2.06 [2.01]
Year dummy for 1999	-0.54 [0.83]	-1.42 [0.44]***	3.72 [1.24]***	-7.09 [0.82]***	-10.82 [1.19]***
Constant	1.94 [0.06]***	3.61 [0.03]***	25.39 [0.11]***	13.48 [0.06]***	68.85 [0.09]***
Subregion dummies	Yes	Yes	Yes	Yes	Yes
Observations	4,240	4,240	4,240	4,240	4,240
Number of subregions	2,302	2,302	2,302	2,302	2,302
R ²	0.35	0.03	0.07	0.61	0.72

Notes: All dependent variables are measured in percentages of total vote. Robust standard errors adjusted for clusters by subregion in brackets.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

6.4 Table 4: 1995 placebo regressions

Read:

- The dependent variables are 1995 electoral outcomes.
- The explanatory variable is NTV availability in 1999.
- The coefficients are much smaller and generally insignificant.

Economic message: NTV availability in 1999 does not strongly predict 1995 electoral outcomes. This supports the assumption that the main 1999 results are not driven by pre-existing political sorting.

6.5 Table 5: Long-term effect in 2003

Read:

- NTV availability in 1999 has persistent positive effects on votes for SPS and Yabloko in 2003.
- It has weaker or mixed effects on KPRF, LDPR, and United Russia.
- It continues to be associated with lower turnout in some specifications.

Economic message: The independent media effect did not disappear immediately after the 1999 election. The 1999 exposure appears to have affected longer-run political preferences for some parties.

TABLE 4—PLACEBO REGRESSIONS FOR THE ELECTIONS OF 1995, AGGREGATE DATA, CROSS SECTION

	Supported by NTV in 1999		Similar coverage by NTV and state TV in 1999			
	Vote for SPS and its predecessor Democratic Russia's Choice in 1995 (liberal)	Vote for Yabloko in 1995 (liberal)	Vote for KPRF in 1995 (communist)	Vote for LDPR in 1995 (nationalist)	Vote for NDR in 1995 (pro-government)	Voter turnout in 1995
NTV availability in 1999	0.60 [0.37]	1.06 [0.83]	-2.33 [3.44]	-2.85 [1.71]	0.12 [1.61]	-3.94 [2.67]
Socioeconomic controls from 1998	Yes	Yes	Yes	Yes	Yes	Yes
Regional dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,686	1,686	1,686	1,686	1,686	1,686
Number of regions	79	79	79	79	79	79
R ²	0.73	0.79	0.73	0.60	0.62	0.67

Notes: All dependent variables are measured in percentages of total vote. The set of socioeconomic controls includes fifth-order polynomial of population, fifth-order polynomial of average wage, number of nurses per capita, number of doctors per capita, and dummy for cities. Robust standard errors adjusted for clusters by region in brackets.

***Significant at the 1 percent level.
**Significant at the 5 percent level.

TABLE 5—LONG-TERM EFFECT OF NTV FOR THE ELECTIONS OF 2003, AGGREGATE DATA

	Supported by NTV in 1999; more time allocated on NTV in 2003; vote for SPS in 2003 (liberal)			Supported by NTV in 1999; similar coverage by NTV and state TV in 2003; vote for Yabloko in 2003 (liberal)		
<i>Panel A</i>						
NTV availability in 1999	1.79 [0.48]***	2.18 [0.53]***	1.06 [0.57]*	1.75 [0.35]***	2.13 [0.43]***	0.38 [0.30]
Electoral controls from 1995	Yes	No	No	Yes	No	No
Electoral controls from 1999	No	No	Yes	No	No	Yes
Socioeconomic controls from 1998	Yes	Yes	Yes	Yes	Yes	Yes
Regional dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,662	1,662	1,662	1,662	1,662	1,662
Number of regions	79	79	79	79	79	79
R ²	0.71	0.67	0.71	0.84	0.79	0.86
Similar coverage by NTV and state TV in 1999 and 2003						
	Vote for KPRF in 2003 (communist)			Vote for LDPR in 2003 (nationalist)		
<i>Panel B</i>						
NTV availability in 1999	-0.71 [1.66]	-1.55 [1.84]	-1.99 [1.55]	-0.5 [1.09]	-0.89 [1.24]	1.42 [1.00]
Electoral controls from 1995	Yes	No	No	Yes	No	No
Electoral controls from 1999	No	No	Yes	No	No	Yes
Socioeconomic controls from 1998	Yes	Yes	Yes	Yes	Yes	Yes
Regional dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,662	1,662	1,662	1,662	1,662	1,662
Number of regions	79	79	79	79	79	79
R ²	0.70	0.63	0.76	0.73	0.68	0.79
Similar coverage by NTV and state TV in 2003; vote for United Russia in 2003 (progovernment, centrist, formed as an alliance of OVR and Unity)						
<i>Panel C</i>						
NTV availability in 1999	-2.95 [3.15]	-3.2 [3.23]	0.28 [3.11]	-6.72 [2.76]**	-8.76 [3.45]**	-2.88 [2.65]
Electoral controls from 1995	Yes	No	No	Yes	No	No
Electoral controls from 1999	No	No	Yes	No	No	Yes
Socioeconomic controls from 1998	Yes	Yes	Yes	Yes	Yes	Yes
Regional dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,662	1,662	1,662	1,662	1,662	1,662
Number of regions	79	79	79	79	79	79
R ²	0.76	0.73	0.79	0.79	0.72	0.82

Notes: All dependent variables are measured in percentages of total vote. Electoral controls from 1995 include vote for KPRF (Communists), vote for Yabloko, vote for NDR (Our Home is Russia), vote for LDPR (Liberal-Democratic Party of Russia), vote for DVR (Democratic Russia's Choice), voter turnout. Electoral controls from 1999 include vote for Unity, vote for OVR, vote for KPRF, vote for SPS, Yabloko, vote for LDPR, and voter turnout. The set of socioeconomic controls includes fifth-order polynomial of population, fifth-order polynomial of average wage, number of nurses per capita, number of doctors per capita, and dummy for cities. Robust standard errors adjusted for clusters by region in brackets.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

6.6 Table 6: Individual-level first stage

Read:

- NTV signal power strongly predicts self-reported watching of NTV.
- The first-stage coefficients are positive in both OLS and probit specifications.

- First-stage exclusion statistics are strong enough to support the IV strategy.

Economic message: The first stage establishes that signal availability is not only a technical measure; it materially affects individual exposure to NTV content.

6.7 Table 7: Individual-level vote effects

Read:

- Instrumented NTV watching reduces the probability of voting for Unity.
- It increases the probability of voting for OVR and SPS.
- It has weaker effects on parties whose coverage was similar across NTV and state television.
- The turnout effect is not as strong at the individual level as in the aggregate analysis.

Economic message: The micro evidence confirms the aggregate mechanism: individuals induced to watch NTV by signal access were more likely to support parties favored by NTV and less likely to support Unity.

	Watched NTV in 1999 (self-reported)			
	OLS	Probit	OLS	Probit
NTV signal power in 1999	0.003 [0.000]***	0.007 [0.001]***	0.002 [0.001]***	0.006 [0.002]***
Gender (one if male)	0.068 [0.035]*	0.218 [0.105]**	0.036 [0.044]	0.122 [0.125]
Age	-0.002 [0.001]**	-0.007 [0.003]**	-0.002 [0.001]	-0.006 [0.004]
Finished high school	0.074 [0.048]	0.195 [0.135]	0.103 [0.056]*	0.272 [0.155]*
Marital status (one if married)	0.03 [0.037]	0.081 [0.107]	0.037 [0.045]	0.099 [0.126]
Consumption index	0.026 [0.013]**	0.084 [0.040]**	0.029 [0.015]*	0.088 [0.046]*
Ln (population), 1998	-0.058 [0.016]***	-0.202 [0.053]***	-0.069 [0.020]***	-0.221 [0.065]***
Ln (average wage), 1998	0.161 [0.046]***	0.584 [0.168]***	0.218 [0.060]***	0.726 [0.217]***
Intention to vote for Unity in 1999			0.08 [0.072]	0.222 [0.201]
Intention to vote for OVR in 1999			0.057 [0.073]	0.141 [0.205]
Intention to vote for SPS in 1999			0.103 [0.085]	0.289 [0.249]
Intention to vote for Yabloko in 1999			0.074 [0.076]	0.209 [0.222]
Intention to vote for KPRF in 1999			0.115 [0.058]**	0.331 [0.162]**
Intention to vote for LDPR in 1999			0.045 [0.118]	0.104 [0.318]
Observations	901	901	656	656
R ²	0.13		0.11	
F-statistics for the exclusion of NTV1999	29.65		13.60	
χ ² statistics for the exclusion of NTV1999		29.9		13.85

Notes: Robust standard errors in brackets. Availability of NTV alone explains 7 percent of variability.

	Opposed by NTV in 1999		Supported by NTV in 1999	
	Vote for Unity in 1999 (centrist, progovernment)	Probit	Vote for OVR in 1999 (centrist, opposition)	Probit
<i>Panel A</i>				
Watched NTV in 1999	-0.831 [0.301]***	-0.139 [0.128]	1.180 [0.477]**	0.135 [0.156]
Marginal effect	-0.26 [0.09]***	-0.05 [0.04]	0.25 [0.14]*	0.02 [0.02]
Controls	Yes	Yes	Yes	Yes
Observations	901	901	901	901
Number of subregions	42	42	42	42
χ ² statistics for the exclusion of NTV1999 in the first stage	34.72		24.79	
	Supported by NTV in 1999			
	Vote for SPS in 1999 (liberal)	Probit	Vote for Yabloko in 1999 (liberal)	Probit
<i>Panel B</i>				
Watched NTV in 1999	1.210 [0.405]***	0.272 [0.149]*	0.467 [0.555]	0.039 [0.178]
Marginal effect	0.24 [0.11]**	0.04 [0.02]*	0.06 [0.08]	0.004 [0.02]
Controls	Yes	Yes	Yes	Yes
Observations	901	901	901	901
Number of subregions	42	42	42	42
χ ² statistics for the exclusion of NTV1999 in the first stage	28.90		30.47	
	Similar coverage by NTV and state TV channels in 1999			
	Vote for KPRF in 1999 (communist)	Probit	Vote for LDPR in 1999 (nationalist)	Probit
<i>Panel C</i>				
Watched NTV in 1999	-0.296 [0.351]	0.052 [0.111]	-0.266 [0.764]	-0.10 [0.140]
Marginal effect	-0.09 [0.11]	0.02 [0.04]	-0.02 [0.07]	-0.01 [0.01]
Controls	Yes	Yes	Yes	Yes
Observations	901	901	901	901
Number of subregions	42	42	42	42
χ ² statistics for the exclusion of NTV1999 in the first stage	30.29		29.91	
	Voter turnout			
	IV probit	Probit		
<i>Panel D</i>				
Watched NTV in 1999	0.219 [0.337]	0.102 [0.115]		
Marginal effect	0.06 [0.08]	0.03 [0.03]		
Controls	Yes	Yes		
Observations	1,148	1,148		
Number of subregions	42	42		
χ ² statistics for the exclusion of NTV1999 in the first stage	35.39			

Notes: Bivariate probit model is used in the IV regressions with “watched NTV” variable from the pre-election survey instrumented by the measure of NTV signal power. Observations are weighted by sample survey weights. Vector of controls includes dummy variables for gender, age, marital status, education, consumption index, logarithms of subregional population, and average wage. Robust standard errors adjusted for clusters by subregion in brackets. Survey weights are applied.

***Significant at the 1 percent level.

**Significant at the 5 percent level.

*Significant at the 10 percent level.

6.8 Table 8: Voting intentions and reported vote

Read:

- Table 8 uses pre-election intentions to separate persuasion from pre-existing preferences.

- NTV still affects reported vote after controlling for intended vote.
- Among respondents who were undecided before the election, NTV affects later vote choices.
- The strongest individual effects are for parties explicitly supported or opposed by NTV.

Economic message: NTV did not merely reach people who already intended to support opposition parties. It shifted voting behavior relative to prior intentions.

Subsample Dependent variable	Opposed by NTV in 1999 Unity in 1999 (progovernment)			Supported by NTV in 1999 OVR in 1999 (centrist, opposition)		
	Excluding undecided		Undecided only	Excluding undecided		Undecided only
	Reported vote	Intention	Reported vote	Reported vote	Intention	Reported vote
<i>Panel A</i>						
Watched NTV in 1999	-0.631 [0.724]	0.073 [0.698]	-1.391 [0.605]**	1.713 [0.133]***	1.508 [0.171]***	0.081 [0.987]
Marginal effect	-0.17 [0.19]	0.01 [0.13]	-0.42 [0.15]***	0.37 [0.03]***	0.41 [0.05]***	0.01 [0.15]
Intention to vote	Yes	No	No	Yes	No	No
Observations	656	786	245	656	786	245
Number of subregions	42	42	42	42	42	42
χ^2 statistics for exclusion of NTV	15.23	14.64	15.40	7.219	17.23	22.05
<i>Panel B</i>						
Subsample Dependent variable	Supported by NTV in 1999			Supported by NTV in 1999		
	SPS in 1999 (liberal)		Undecided only	Yabloko in 1999 (liberal)		Undecided only
	Excluding undecided		Reported vote	Excluding undecided		Reported vote
	Reported vote	Intention	Reported vote	Reported vote	Intention	Reported vote
Watched NTV in 1999	1.808 [0.161]***	-0.329 [0.472]	1.033 [0.389]***	-0.164 [0.856]	1.103 [0.316]***	1.040 [0.644]
Marginal effect	0.28 [0.03]***	-0.05 [0.07]	0.17 [0.09]**	-0.01 [0.06]	0.24 [0.10]**	0.14 [0.12]
Intention to vote	Yes	No	No	Yes	No	No
Observations	656	786	245	656	786	245
Number of subregions	42	42	42	42	42	42
χ^2 statistics for exclusion of NTV	12.35	15.11	20.20	13.45	16.68	20.04

(Continued)

Table 8, Panels A–B

Subsample Dependent variable	Similar coverage by NTV and state TV channels in 1999					
	KPRF in 1999 (communist)			LDPR in 1999 (nationalist)		
	Excluding undecided		Undecided only	Excluding undecided		Undecided only
	Reported vote	Intention	Reported vote	Reported vote	Intention	Reported vote
<i>Panel C</i>						
Watched NTV in 1999	-0.746 [0.412]*	-0.713 [0.874]	0.804 [0.757]	-0.584 [1.020]	-0.347 [0.794]	1.075 [0.796]
Marginal effect	-0.16 [0.10]	-0.22 [0.26]	0.21 [0.20]	-0.04 [0.10]	-0.03 [0.08]	0.13 [0.17]
Intention to vote	Yes	No	No	Yes	No	No
Observations	656	786	245	656	786	245
Number of subregions	42	42	42	42	42	42
χ^2 statistics for exclusion of NTV	15.04	9.040	22.68	14.11	14.33	22.49
<i>Panel D</i>						
Subsample Dependent variable	Voter turnout in 1999			Voter turnout in 1999		
	Excluding undecided		Undecided only	Excluding undecided		Undecided only
	Reported vote		Reported vote	Reported vote		Reported vote
	Reported vote	Intention	Reported vote	Reported vote	Intention	Reported vote
Watched NTV in 1999	-0.609 [0.525]	0.313 [0.449]	0.378 [0.453]			
Marginal effect	-0.12 [0.12]	0.05 [0.08]	0.13 [0.15]			
Intention to vote	Yes	No	No			
Observations	764	1,105	384			
Number of subregions	42	42	42			
χ^2 statistics for exclusion of NTV	15.33	30.86	23.27			

Notes: Bivariate probit model. Watched NTV variable from the postelection survey instrumented by the measure of NTV signal power. In columns marked "undecided only" only respondents who did not report their intention to vote in the preelection survey are included in the sample. Observations are weighted by sample survey weights. All specifications include the following controls: gender, age, marital status, education, consumption index, logarithm of subregional population, and wage. Controls for intention to vote include dummy variables for intention to vote for six major parties. Robust standard errors adjusted for clusters by subregion in brackets.
***Significant at the 1 percent level.

Table 8, Panels C–D

7 Short summary

- The paper estimates the effect of independent television on voting in the 1999 Russian parliamentary election.
- NTV was the only national television channel independent from the government.
- The key treatment is geographic availability of NTV signal.
- NTV opposed the pro-government Unity party and supported OVR, SPS, and Yabloko.
- NTV availability reduced Unity's vote share and increased support for opposition and liberal parties.
- NTV availability also reduced turnout.
- The aggregate implied effects are large: Unity lost about a quarter of its voters and opposition parties increased support by about 60 percent.
- Placebo tests using 1995 outcomes support the identifying assumption.
- Panel evidence and long-term 2003 evidence reinforce the main interpretation.
- Individual-level IV results show that signal power predicts NTV watching and that NTV watching changes reported voting behavior.
- Persuasion-rate calculations show that the negative message against Unity was especially powerful.
- The paper's central lesson is that independent media can have very large effects in new democracies with weak party systems and concentrated media markets.

8 Conclusion

8.1 Core conclusion

Enikolopov, Petrova, and Zhuravskaya show that independent television had a large effect on the 1999 Russian parliamentary election. NTV reduced support for the government party, increased support for opposition parties, and lowered turnout. The effect is visible in aggregate election data and in individual survey data.

8.2 Economic intuition

The intuition is that voters in weakly institutionalized democracies may have limited information and weak party attachments. When the state dominates most media outlets, access to an independent national channel can substantially change the information set of voters. Because NTV's coverage differed sharply from state television, viewers learned different political messages and updated their voting decisions.

8.3 Incremental contribution takeaway

The paper's incremental contribution is to show that media persuasion can be very large when democratic institutions are weak, party brands are unstable, and the media environment is concentrated. It also shows that the content of persuasion matters: negative messages against a government party can have much larger effects than positive messages supporting opposition parties.

8.4 Takeaway

Independent media are not merely an institutional detail; they can materially change electoral outcomes. This helps explain why governments in hybrid regimes seek to control television and why access to alternative media sources can be central to political competition.